

BMJ Open Type I hybrid effectiveness-implementation randomised controlled trial to address intergenerational impact of war trauma and resilience among second-generation refugee children in the USA: Resettled Refugee Families for Healing (RRF4H) study protocol

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To cite: Tutlam NT, Liyew TW, Betancourt TS, *et al.* Type I hybrid effectiveness-implementation randomised controlled trial to address intergenerational impact of war trauma and resilience among second-generation refugee children in the USA: Resettled Refugee Families for Healing (RRF4H) study protocol. *BMJ Open* 2025;**15**:e108824. doi:10.1136/bmjopen-2025-108824

► Prepublication history and additional supplemental material for this paper are available online. To view these files, please visit the journal online (<https://doi.org/10.1136/bmjopen-2025-108824>).

Received 02 August 2025
Accepted 12 September 2025



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ABSTRACT

Introduction Children from refugee families resettled in the USA face higher risks of serious mental health challenges compared with their native-born peers. Research shows that refugee youth in high-income countries frequently suffer from trauma-associated disorders such as post-traumatic stress disorder (PTSD), depression and anxiety. The high prevalence of trauma-associated mental health problems among these youth may be attributed to their own trauma exposure, especially if born in conflict zones, and post-resettlement challenges like poverty, acculturation difficulties, racism and discrimination. However, they may also suffer from the effects of intergenerational trauma, where parental war trauma impacts them. This study aims to adapt and test an intervention addressing intergenerational trauma-related emotional and behavioural health outcomes among US-born children of refugee parents in Omaha and Lincoln, Nebraska.

Methods and analysis This is a two-arm cluster randomised type I hybrid effectiveness-implementation trial. Guided by the Social Action and Family Systems theories and applying them to the intergenerational transmission of trauma framework, the combination intervention consists of family strengthening model delivered through multiple family groups+peer mentoring programme called TeenAge Health Consultants (TAHC) adapted for delivery in virtual environment (Virtual TAHC). A total of 154 US-born adolescents of parent resettled as refugees (77 per study arm), ages 14–17 and at least one biological parent per youth (dyads) will be recruited from four comparable communities utilising community-based participatory research approach and randomised to usual care or intervention group. The intervention will be implemented for up to 16 weeks, with assessments at baseline, after intervention completion and 6 months follow-up. To determine study feasibility, we will use binary metrics of participant enrolment of 70% or more and retention of 80% or more at 12 months. To

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ This study adapts and combines two evidence-based interventions consisting of a family strengthening model delivered through multiple family groups and a peer mentoring programme called Teenage Health Consultants.
- ⇒ This adaptation of the peer mentorship programme from an in-person delivery to virtual format using interactive animations and avatars will allow for tailoring of the intervention to unique subpopulations.
- ⇒ This type I hybrid effectiveness-implementation design will allow us to test specific implementation strategies (audit and feedback; build a coalition; conduct educational meetings; and develop educational materials) to identify relevant implementation strategies.
- ⇒ The study employs a community-based participatory research approach for both adaptation and participant recruitment.
- ⇒ Resettled refugee communities in the USA come from very diverse cultural, religious and linguistic backgrounds, and it may not be possible to accommodate the needs of all subgroups in the intervention adaptation process.

assess study acceptability, we will determine participant satisfaction with the study based on the Client Satisfaction Questionnaire (CSQ-8). To maximise rigour, our analyses will follow an intention-to-treat (ITT) approach. For primary inferential analyses, we will fit two-level generalised linear mixed models to continuous primary outcomes. The models will include fixed effects for study arm, time and their interaction terms. We will perform time-averaged comparisons of post-baseline repeatedly measured observations across study arms to examine intervention effects over the duration of the postintervention study period. To delineate barriers and facilitators to

implementation and implementation strategies, we will apply a more integrative approach, using both inductive and deductive approaches guided by the grounded theory and integrative theory that combines both deductive and inductive approaches. Finally, we will integrate findings from the quantitative and qualitative analysis to provide additional explanation and context for our quantitative findings.

Ethics and dissemination Voluntary written informed assent and consent will be obtained from all participants, adolescents and their parents, respectively. All study procedures received approval from Washington University in St. Louis Institutional Review Board (IRB #202307081).

Study findings will be disseminated through publications in scientific journals and presentations at national and international conferences. We also plan to provide community education about the study through a dissemination conference at the end of the study.

Trial registration number NCT06176638.

INTRODUCTION

War trauma has devastating direct and indirect effects on human health and development, especially on the mental health of populations affected.¹ Refugees resettled in the USA often come from war-ravaged countries where they experienced a great deal of war-related traumatic events such as sexual assault/rape, witnessing violent death of loved ones, forced prostitution, torture, political violence, genocide, combat situation/assault, famine, displacement and abduction/kidnapping.² Consequently, these communities have higher rates of trauma-associated mental illness.^{3–7} Indeed, previous studies have found high prevalence of trauma-associated disorders including anxiety (3%–96%), depression (3%–30%) and post-traumatic stress disorder (PTSD) (3%–94%) in populations affected by armed conflict resettled in high-income countries.^{3 4 8–14}

Importantly, children from resettled refugee families have elevated risk of developing serious mental health challenges compared with their US native peers.^{3 4 15 16} For example, a study by Tutlam and colleagues among US-born children of parents resettled as refugees, here after referred to as US-born children, in Omaha and Lincoln, Nebraska (sites of the current study) and Nashville, Tennessee, indicated that prevalence of trauma-associated disorders was comparatively high: 11.8% for aggressive behaviour; 7.9% for antisocial behaviour; 7.9% for anxiety disorder; 6.6% for depressive disorder; 10.5% for PTSD and 21.0% for meeting diagnostic criteria for any disorder.¹⁷ Other studies have also found that refugee youth living in high-income countries have high prevalence of PTSD, ranging from 19% to 54%, and depression ranging from 3% to 30%.⁴ Additionally, youth belonging to these vulnerable communities have high rates of substance use, school dropout, incarcerations, youth violence including gang-related homicides and suicides, which have emerged as serious public health threats in recent years.^{18–28} While the high prevalence of mental health problems among these youth may be due to their own exposure to violence and post-resettlement factors such as poverty, acculturation difficulties, racism

and discrimination,^{29–34} they may also be shaped by the impact of intergenerational transmission of trauma. This phenomenon is defined as ‘the process in which trauma that happened to the first generation is passed on to the second generation’.^{35 36}

The effects of intergenerational transmission of trauma manifest in children affected in the form of emotional and behavioural conditions such as aggression, antisocial behaviour, anxiety, depression, PTSD or somatisation.^{37–39} These are debilitating conditions with serious public health consequences. Most intergenerational transmission of trauma studies have been conducted among children of Holocaust survivors and some refugee populations from Asia⁵; thus, it is necessary to investigate this phenomenon among other refugee children, particularly from sub-Saharan Africa (SSA), where a large proportion of refugees and victims of other adversities come from. The studies that have attempted to address this question in other populations have found an association between parental PTSD and the development of PTSD among children of Cambodian, South Sudanese and Vietnamese refugees resettled in the USA,^{7 17 40 41} torture victims,⁴² children of Vietnam war veterans in the USA and other settings like Australia,^{37 43–48} children of the 1994 Rwandan genocide survivors,^{49–51} and children in the Nurses’ Health Study II and Growing Up Today Study.⁵² Similarly, a study by Tutlam *et al* showed that the joint effects of maternal trauma and anxiety, trauma and depression, and trauma and PTSD increased the odds of meeting diagnostic criteria for depression by 36%–47% among US-born children.⁵³

Crucially, there is a huge gap that studies focused on children belonging to resettled refugee communities do not address. Specifically, studies in this population seldom distinguish between refugee children who themselves experienced trauma first-hand and those that were born after their parents resettle in the USA. This distinction is crucial given the potential intergenerational impact of war trauma in this population. Additionally, because of the unique and brutal nature of the traumatic events experienced in war, and the devastating impact of war trauma not only on individuals directly affected but possibly on subsequent generations, and the significant number of people affected by conflict resettled in the USA, it is critical to examine the intergenerational impact of war trauma on children of parents affected by war.

To examine and address the complex factors that may contribute to the suboptimal mental health of US-born children, a multilevel intervention that can address the mental health challenges at different levels is needed.^{19 54} Necessarily, this intervention should have components that target the hypothesised mechanisms of transmission outlined in figure 1. Accordingly, this type I hybrid effectiveness-implementation trial⁵⁵ aims to test a combination intervention designed to improve intergenerational trauma-related mental health symptoms among US-born children and their biological parents. We will use the Exploration, Preparation, Implementation,

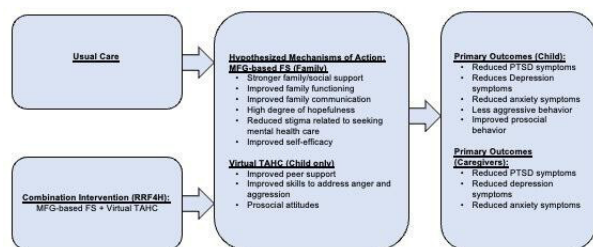


Figure 1 RRF4H study conceptual model. FS, family strengthening; MFG, multiple family group; PTSD, post-traumatic stress disorder; RRF4H, Resettled Refugee Families for Healing; TAHC, TeenAge Health Consultants.

Sustainment (EPIS) framework to guide the implementation and adaptation process.⁵⁶ Moreover, guided by the Social Action⁵⁷ and Family Systems⁵⁸ theories, and applying them to the intergenerational transmission of trauma framework, the combination intervention consists of family strengthening (FS) model delivered through multiple family groups (MFG)+peer mentoring programme called TeenAge Health Consultants (TAHC) adapted for delivery in virtual environment (Virtual TAHC). The study has the following specific aims:

Aim 1

To systematically adapt an evidence-based FS model delivered through MFG combined with a peer mentoring programme (Virtual TAHC).

Hypothesis

The combination intervention will be deemed feasible and acceptable among resettled refugee communities and inform refinement of implementation strategies prior to larger trial to test effectiveness.

Aim 2

To assess preliminary short-term and long-term impact of the combination intervention (Resettled Refugee Families for Healing, RRF4H) on behavioural emotional disorders (aggressive behaviour, antisocial behaviours, anxiety, depression and PTSD) related to intergenerational trauma among US-born children in the trial.

Hypothesis

Compared with children in the usual care arm, participants in the RRF4H intervention group will show improvements in primary mental health outcomes.

Aim 3

Using mixed methods and applying the EPIS framework, examine implementation strategies, facilitators and barriers of the RRF4H intervention.

Hypothesis

We hypothesise that the use of implementation strategies guided by the EPIS framework will enhance implementation and sustainment of the intervention.

Theoretical framework

The intervention in response to this complex chain of events leading to the high level of trauma-associated mental disorders among US-born children is guided by the Social Action⁵⁷ and Family Systems⁵⁸ theories, which take into account the link between protective parenting practices and family processes, as well as contextual circumstances and supports in order to address youth mental disorders. Specifically, Social Action theory posits that a person's contextual influences (action contexts) impact their self-change processes (ie, social support, health expectancies, self-efficacy), which in turn impact their actual health behaviour or action.⁵⁹ As such, factors tied to immigration status and poverty (eg, trauma, economic and social stress, and discrimination) may undermine parenting (eg, family organisation, discipline practices, family connectedness, support and communication) and contribute to serious youth mental disorders.^{59–64} In collaboration with parents and service providers in the USA, MFG delivered FS that was developed and guided by a protocol which encouraged transparency of the evidence base for families and provided an 'easy to remember' means of organising existing science for lay MFG facilitators. Specifically, four broad conceptual categories are the family-level targets for FS (FS): Rules, Responsibility, Relationships and Respectful communication (4Rs). Stress and Social support (2Ss) were added as these impact service engagement and outcomes for children raised within poverty-impacted families.^{65–72}

Figure 1 shows our conceptual model depicting how the combined intervention is expected to improve mental health outcomes among both participating youth and their biological parents compared with those who only receive usual care.

Additionally, in line with National Institute of Mental Health (NIMH) experimental therapeutic approach, using directed acyclic graph (figure 2), we show our hypothesised mechanism of action through which the combined intervention potentially acts to improve mental health outcomes among participants. For example, the RRF4H is expected to improve peer support among the youth and improve degree of hopefulness among both the youth and their parents, which can directly improve mental health symptoms such as depression among both the adolescents and their parents. Improvement of peer support may also directly increase prosocial attitudes among youth in particular, and that, in turn, is also expected to improve their mental health outcomes.

We apply the Social Action⁵⁷ and Family Systems⁵⁸ theories to the intergenerational transmission of trauma framework to target factors that increase the risk of mental disorders among US-born children. Intergenerational transmission of trauma is defined as 'the process in which trauma that happened to the first generation is passed on to the second generation'.^{35 36} The trauma in the children whose parents were exposed to trauma manifests in the form of psychological and behavioural disorders such

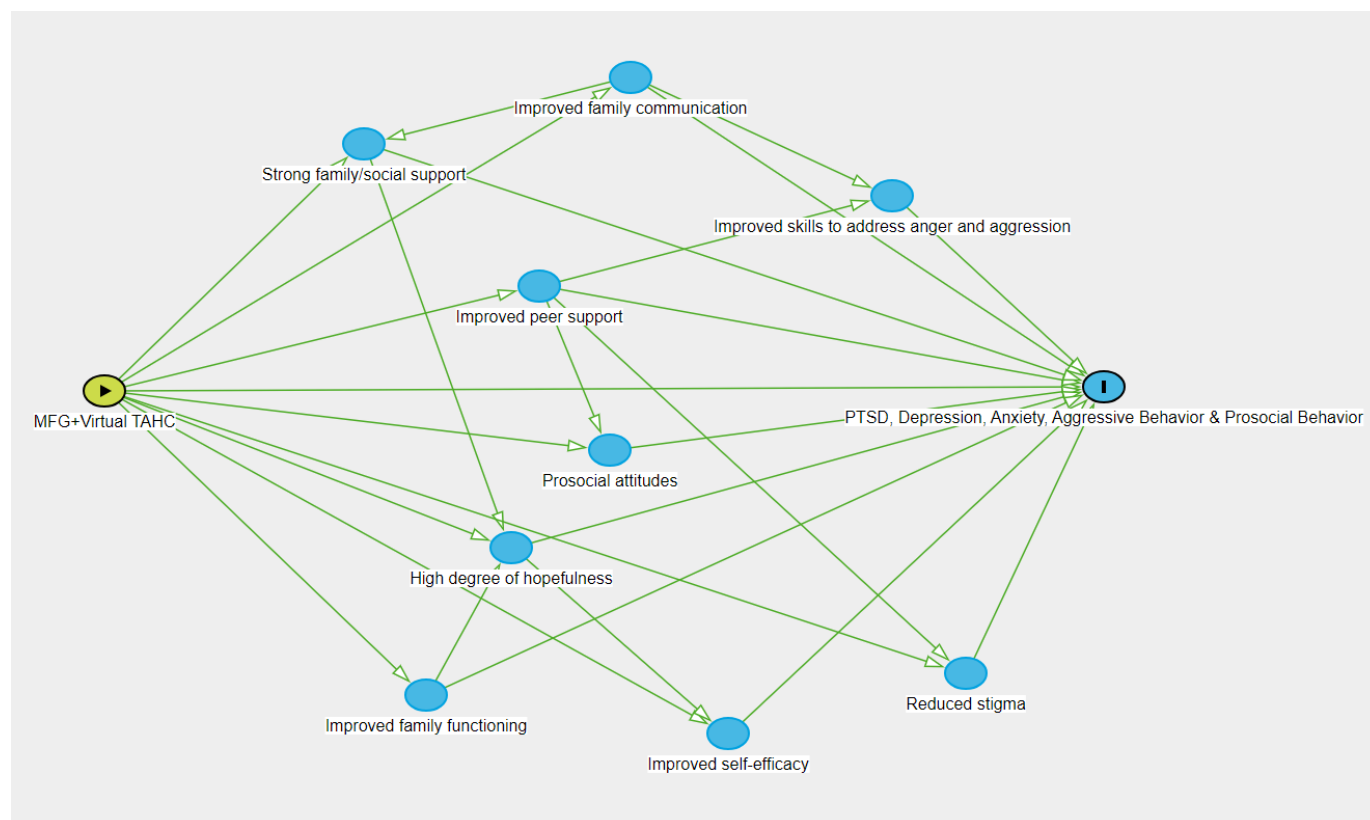


Figure 2 Directed acyclic graph representation of the hypothesised mechanisms of action. MFG, multiple family group; PTSD, post-traumatic stress disorder; TAHC, TeenAge Health Consultants.

as aggression, antisocial behaviours, anxiety, depression, PTSD and somatisation.³⁸

Potential mechanisms of transmission of intergenerational trauma (ITT)

It is well established that war trauma has significant impact on the mental health of population affected.^{1 73} With respect to intergenerational transmission of trauma, as shown in figure 3, we postulate that parental trauma or the impact of trauma leads to the emotional and

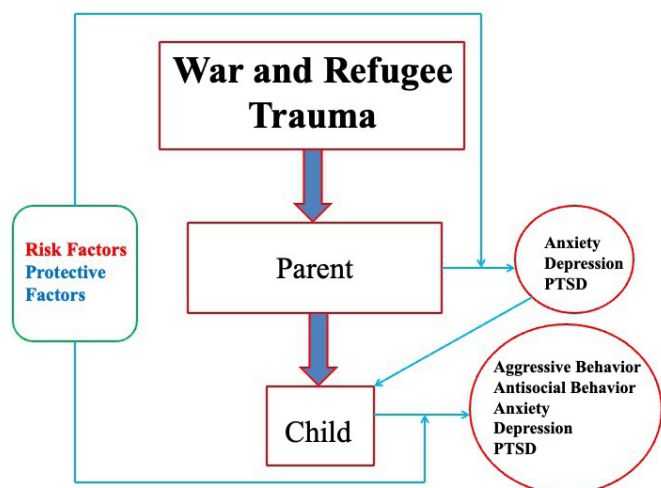


Figure 3 Intergenerational transmission of trauma conceptual framework. PTSD, post-traumatic stress disorder

behavioural problems among their US-born children through a number of mechanisms that have been proposed by Kellermann.⁷⁴ These include (1) psychodynamic relational models; (2) sociocultural and socialisation models; (3) family systems and communications models and (4) biological or genetic models.⁷⁴ These models, broadly speaking, fall under two major mechanisms of transmission: (1) direct transmission, including psychodynamic and biological models, where symptoms of traumatic effects such as PTSD are directly transmitted to the child with direct impact on the child. In this case, the main factors that are thought to be transmitted include displacement of emotions unconsciously from parent to child and heightened vulnerability to PTSD that is inherited and (2) indirect transmission, where children are affected by negative impacts of PTSD in their parents. Specifically, in the case of socialisation the main factor that is transmitted is parenting and modelling and enmeshment, where parents become so obsessive with their children to the point their children become snared in their emotional problems. Additionally, studies have suggested that the mental health of children is affected through various mechanisms including insecure attachment, decreased family functioning, dysfunctional communication style, accumulation of family stressors and stigma.^{40 75} The multiple potential mechanisms of action identified in figure 1 will allow us to target multiple factors that lead to increased risk of mental disorders

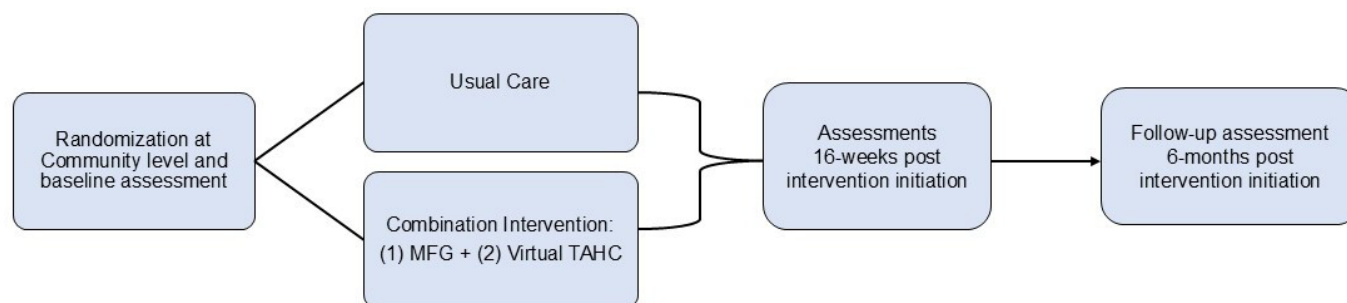


Figure 4 RRF4H study design. MFG, multiple family group; RRF4H, Resettled Refugee Families for Healing; TAHC, TeenAge Health Consultants

both among the adolescents and their parents. Therefore, this intervention is best suited for addressing the multiple adverse mental health outcomes in this unique population of youth and their families.

METHODS

Study design

The RRF4H study is a type I hybrid effectiveness-implementation trial. Using a two-arm cluster randomised controlled trial (RCT), we plan to recruit 154 adolescents (77 per study arm), ages 14–17 and at least one biological parent per youth from resettled refugee communities. We selected this age range because this age group will allow us to better understand developmental mechanisms of trauma transmission. Moreover, addressing trauma during adolescence can mitigate long-term adverse mental health outcomes.⁷⁶ Participants will be randomised to one of two treatment conditions at the community level: (1) usual care and (2) combination intervention consisting of MFG+Virtual TAHC peer mentoring programme. Data will be collected at baseline (preintervention), 16 weeks and 6 months postintervention (figure 4). We used the Standard Protocol Items: Recommendations for Interventional Trials (SPIRIT) reporting guidelines for this study.⁷⁷

RRF4H study arms

Control arm (usual care)

All adolescents recruited into the RRF4H study will receive the usual mental health counselling provided through their school counsellors. Schools in Omaha and Lincoln, Nebraska provide individual and small group counselling to help students ‘resolve conflict, practise skills and think through their actions’.^{78 79} There is no structured curriculum for the group counselling programmes but they are available to all students as needed. The Lincoln Public School District also offers additional resources on specific topics such as trauma, depression and anxiety in children and adolescents, as well as alcohol and substance use in families and provides appropriate referrals for those in need. Additionally, through organisations such as the International Council for Refugees and Immigrants based in Omaha, refugee youth aged 7–18 years can receive educational and social support programmes,

after-school STEM (Science, Technology, Engineering, and Mathematics) clubs and one-on-one peer mentoring.⁸⁰ Through the New Life Family Alliance, in addition to after-school programme and the boys and girls basketball programme offered twice a month, youth are connected to youth-serving agencies that can help them effectively and successfully develop and take advantage of opportunities available to them in the state of Nebraska.⁸¹

Treatment arm (RRF4H Intervention)

The treatment arm of the RRF4H study is an innovative combination intervention that builds on the usual care consisting of (1) an FS model delivered through MFG model⁸² and (2) a peer-mentoring programme called Virtual TAHC adapted for delivery in virtual environment. The combination intervention components target factors in our hypothesised mechanism of change such as communication, relationship and social support network development to assist with parenting and stress management, stigma reduction, peer support and prosocial attitudes. These are described below:

Multiple family groups

This model was developed by MM (coauthor) in 1995 to address the mental health needs of minority children and families in inner cities^{82 83} and has been successfully implemented in projects led by FMS (coauthor), in SSA population. The basic idea is that if children and their parents discuss and address common problems together and build a supportive network, this could be an efficient and effective means of providing mental health services to children and their families.⁸² We believe the MFG component will ultimately help address mental health outcomes, family functioning and communication among both adolescents and parents. This is because the MFG approach provides a social support mechanism and strengthens family relationships by allowing families to share common experiences as well as effective strategies for addressing difficult issues and focuses on reducing stigma and normalising common experiences.⁸⁴ Indeed, stigma has been identified as one of the mechanisms through which intergenerational trauma is perpetuated⁷⁵ and the MFG is designed to address this problem. Moreover, the MFG is deemed appropriate for addressing this problem with complex layers of factors in this particular

population because the family-based method is congruent with the target population's cultural practices that value communal approach to problem-solving and raising children.⁸⁴ The MFG has been used in different situations to address various conditions including disruptive behaviour disorders,^{85 86} PTSD treatment among Bosnian refugees⁸⁷ and helping reduce problem behaviours (internalising and externalising) among incarcerated adolescent sex offenders.⁸⁸ Additionally, this model has been applied to treat adolescents with substance use disorder, which demonstrated that adolescents benefited from involvement of their families in their treatment process.⁸⁹ Moreover, this model meaningfully improved child and parent outcomes when it was implemented on multistressed families.⁹⁰ Based on factors we articulated previously, our target population, both the adolescent and their parents, fits this description of multistressed families. Previous studies have found that among these groups, empowerment, family connectedness, perception of caring adults in the community and quality after-school activities were protective against depressive symptoms.⁹¹ Similarly, a family-based home-visiting preventive intervention implemented by TSB (coauthor) has been found to be feasible and acceptable with promising results for improving mental health and family functioning.⁹² All that said, there remains a gap that these interventions have not addressed, which we would be able to address with our study. Specifically, there are behavioural issues such as substance use, antisocial behaviour, aggressive behaviour, lack of problem-solving skills and violence prevention education opportunities that may not be addressed by MFG alone. Because these are prevalent serious mental health problems among US-born children, the adapted peer mentoring programme is a crucial component of the intervention.

Intervention delivery

The intervention will be delivered to families (both the youth and at least one biological parent) in groups of 8–10 families per group. These will be facilitated by individuals outside the research team, recruited from the community and trained on specific MFG content areas. On completion of the training, we will assess the knowledge level of the facilitators using a knowledge and skills assessment test. A cut-off score of 80% and above will be used to determine mastery of materials. In addition to reimbursement for their time, facilitators will receive a certificate of completion of MFG Facilitator Training. The facilitators will be trained by the PI who has been trained by the developer of the programme (MM).

Adapted peer mentoring intervention (Virtual TAHC)

Youth enrolled in the intervention condition will also receive a peer mentoring programme adapted to be delivered in a virtual environment. This is a peer-led evidence-informed programme that allows younger students to talk with peers about important issues that they face, gain role models and identify positive social norms from older

peers. It provides age-appropriate lessons and is delivered in a structured fashion. The curriculum consists of sixteen 50 min lessons spread out over two academic years. Topics delivered in the curriculum include substance use, positive outlook on life, forming a positive self-concept, decision making and problem solving, coping with depression, bullying and social media. To optimise the potential impact of the programme, we propose modification to deliver the 16 lessons over an abbreviated period of time determined during the adaptation process. This is also the same length of time required for the MFG component. More importantly, because of widespread use of technology by youth and the ease and cost-effectiveness of delivering interventions in a consistent manner to youth through technology, technology-based interventions have been found to be ideal for delivering evidence-based interventions and provide promising results in prevention of risky behaviour such as substance use disorder among youth.⁹³ In this regard, we propose to adapt the TAHC component of the combination intervention to be delivered as an interactive module in a virtual environment. The adapted version *will be called Virtual TeenAge Health Consultants (Virtual TAHC)*. The application will be designed such that the role-plays can be performed through interactive animations and avatars, facilitated by peer mentors who can engage the participants in this environment, in the same manner multiple players from different locations play video games. We believe that given the affinity of the youth for technology, this will provide an ideal opportunity to deliver programme that can be scaled up or down as the situation warrants. Importantly, this virtual environment also will have built-in tools that will allow for near real-time assessment of knowledge change. Additionally, given unforeseen challenges such as those imposed by the COVID-19 pandemic, this adaptation will provide an easy and cost-effective way to deliver the programme without the need for in-person gathering. The technology is currently available to design and implement this adaptation.

Intervention delivery

TAHC provides age-appropriate lessons and is delivered in a structured fashion. To improve retention of knowledge, as part of the adaptation, we propose to deliver the intervention over a shorter period of time, determined during the adaptation process. To achieve behaviour change, it is important to set a goal of no less than eight sessions. This is the number of sessions considered necessary to achieve meaningful behaviour change.⁹⁴ We will align the number of sessions with the MFG sessions. This intervention will be delivered to only the youth as a digital intervention. The intervention will be facilitated by upper-class students called TAHC leaders. These TAHC leaders will be identified by their teachers or community leaders (sponsors) in their respective schools and will undergo a 2-day training by the research team and the consultant who will help with the adaptation of the programme into a digital intervention. The training will involve learning contents in the

TAHC curriculum (listed above) and how to navigate the digital intervention. TAHC leaders will receive incentives including gift cards, achievement certificates and extra credit that will go towards their community service hours. The TAHC leaders will deliver the intervention in virtual group sessions (ratio 1:13). Specifically, using avatars that are designed to reflect the target population, we will simulate different scenarios that youth engage in that could be potentially harmful to them, and the facilitators will engage participants in the virtual environment and guide the participants in discussions on how to respond to these kinds of scenarios with the goal of encouraging prosocial behaviours and reducing maladaptive behaviour. To improve the intervention delivery strategy by TAHC leaders, we will employ audit and feedback as an intervention strategy,⁹⁵ where they will receive guidance and feedback on how to improve the delivery of the intervention after every session.

Intervention adaptation (aim 1)

For the MFG component, although the intervention was developed for low-income communities in the USA, which are similar to where many of the refugees are settled in Omaha and Lincoln, Nebraska, we will need to adapt it to fit the needs of the new target population of resettled refugee families, thereby fostering bidirectional learning through our work in the global north and global south. We will use the same procedures used by MM and FMS to adapt the MFG to the SSA context.^{96 97} To cross-culturally adapt Virtual TAHC and the combination intervention, we will employ the Assessment, Decisions, Administration, Production, Topical experts, Integration, Training staff and Testing (ADAPT-ITT) model.⁹⁸ The ADAPT-ITT model is a framework designed to provide a systematic method for adaptation of evidence-based interventions for prevention. It consists of eight phases: (1) assessment through focus groups and key informant interviews; (2) decisions on how to adapt the intervention; (3) administration of the intervention to test it with the target population; (4) production of first draft of adapted manual and development of quality assurance and process measures; (5) identification of topical experts to review session contents and provide feedback; (6) integration of feedback from topical experts into the second draft of adapted manual and finalise quantitative assessment battery; (7) training intervention facilitators, supervisors and research assistants and (8) prepilot testing the adapted intervention. Specifically, the adaptation process will involve meeting stakeholders (phase 1); reviewing the manual by the team, including community members (phase 2); meeting with community stakeholders (phase 3); finalising curriculum content (phase 4) and developing illustrations for the manual (phase 5). These adaptation steps will be integrated within the EPIS framework⁵⁶ that is used to guide the adaptation and the implementation process. Specifically, EPIS' exploration phase will guide the assessment, decision-making and administration phases of the adaptation to understand

the problems and identify the needs of the intervention. Similarly, the preparation phase where intervention planning and development takes place in consultation with stakeholders will guide the production, consultation with topical experts and integration of their feedback into the adapted intervention. Finally, the implementation phase of the EPIS framework will guide training of the implementers and testing of the adapted intervention. The overall intervention will be adapted for language, cultural and religious sensitivities of the target population and as part of the adaptation process, we will assess barriers and facilitators of the adaptation in this setting and population from those participating in the intervention. Consistent with the community-based participatory research (CBPR)^{99 100} approach that we are employing in this study, we will identify relevant implementation strategies and carefully specify them based on guidance from Proctor *et al.*¹⁰¹ We anticipate employing the following implementation strategies: (1) audit and feedback; (2) build a coalition; (3) conduct educational meetings and (4) develop educational materials.⁹⁵ These implementation strategies are aimed at enhancing important implementation outcomes proposed by Proctor *et al.*,¹⁰² namely acceptability, feasibility and fidelity. As part of the intervention adaptation, if warranted, adaptations will be made to these implementation strategies and any modifications will be carefully documented as proposed by Haley *et al.*¹⁰³ and Miller *et al.*¹⁰⁴

Study setting

The study will be implemented among resettled refugee families from SSA) residing in communities in Omaha and Lincoln, Nebraska. Community is defined as geographically designated areas of Omaha and the City of Lincoln, Nebraska. The study will target four areas (communities) with significant populations of resettled refugees: (1) North Omaha; (2) South Omaha; (3) West Omaha and (4) City of Lincoln, Nebraska. In 2016, Omaha, Nebraska had the highest number of refugees per capita in the USA.¹⁰⁵ It is home to many resettled refugees including approximately 10 000 from South Sudan,²² and a growing population of refugees from Somalia.¹⁰⁶

Study participants

The target population for this study is US-born children of resettled refugees and their biological parents (dyads), and peer mentors who will deliver the Virtual TAHC programme. US-born children inclusion criteria include: (1) child born in the USA; (2) ages 14–17 years; (3) in the 9th–12th grades and (4) enrolled in one of the schools in Omaha and Lincoln, Nebraska area schools. All eligible children from the same family will be enrolled and assigned to the same study arm. Peer mentors inclusion criteria: (1) peer mentors (n=6; ratio of 1:13) must be students in upper class (11th or 12th grade); (2) willing and able to serve as good role models and (3) nominated by community leaders or teachers. Parent inclusion criteria: (1) resettled in the USA as a refugee;

(2) must be a biological parent of a child between the ages of 14–17 years born in the USA and (3) must be ≥ 30 years of age. Exclusion criteria for all participants: cannot understand the study procedures and/or participants' rights during the informed consent process; unwilling or unable to commit to completing the study; and present with emergency needs (eg, hospitalisation) in which case needed care will be sought rather than study participation. All youth will be screened for severe psychiatric illness using the Quick Structured Clinical Interview for Diagnostic and Statistical Manual of Mental Disorders - 5 (DSM-5)¹⁰⁷ and those who screen positive for psychosis or acute suicidality will be referred to treatment rather than project participation. Additionally, we will use the screening form judgement of the trained research assistants to identify those who are unable to comprehend the informed consent/assent process (assent and consent forms available as online supplemental documents) and study expectations or unwilling to complete study activities will not be enrolled in the study. The study start date was 1 June 2024, with an anticipated study completion date of

31 December 2026. Recruitment commenced at the start of the study period through the end of September 2024. The timeline of the project activities is listed in [table 1](#).

Randomisation of participants

To minimise contamination, we will randomise the communities of recruitment. Randomisation will take place at the community level (cluster randomisation) so that all participants in a given community will receive the same intervention. Community will be defined as geographically designated areas of Omaha, and the City of Lincoln, Nebraska. The study will target four areas (communities) with significant populations of resettled refugees: (1) North Omaha; (2) South Omaha; (3) West Omaha and (4) City of Lincoln, Nebraska. The communities from where participants will be recruited are comparable based on CDC's social vulnerability index (SVI).¹⁰⁸ The SVI ranks the vulnerability of communities at the census tract level on a scale ranging from 0 to 1 with 0 being the least vulnerable and 1 the highest vulnerable communities. The index is generated based on 16 social

Table 1 Quarterly research activities timeline

Activity	Year 1				Year 2				Year 3				Year 4			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Seek IRB approval	x															
Meeting church and community leaders	x															
Prepare study site	x	x														
Systematic adaptation of TAHC and combination intervention		x	x													
Conduct randomisation of communities				x												
Recruit and train research assistants			x	x												
Recruit participants			x	x												
Conduct baseline assessment				x	x											
Intervention delivery treatment group parents and youth (MFG); (Virtual TAHC) youth only						x	x	x	x							
Conduct assessment 16 weeks postintervention initiation							x	x	x	x						
Conduct 6-month follow-up assessment											x	x	x	x		
Qualitative interviews, focus groups													x	x		
Data entry, cleaning, analysis					x	x	x	x	x	x	x	x	x	x	x	x
Dissemination: ongoing community engagement,	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Study protocol manuscript,				x	x											
Study progress reports	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Preparation of manuscripts based on data from the study						x	x	x					x	x	x	x
Write and submit R01									x	x	x	x	x	x	x	x

IRB, institutional review board; MFG, multiple family group; TAHC, TeenAge Health Consultants.

factors including poverty, education, family characteristics, housing and language ability. As this study involves intervention where participants will be engaged in activities that are obviously designed to affect the outcome of the study, delivered by facilitators and peer mentors, blinding is not possible.

Recruitment strategies

The proposed research project will employ CBPR approach,^{99 100} building on existing strong relationships and structures which have been established through years of partnership and collaboration between the PI and community organisations working directly with resettled refugee communities. Specifically, to enhance our outreach and access to all refugee communities in Omaha and Lincoln, Nebraska, we are partnering with the Mental Health Subcommittee of the Omaha Refugee Task Force, that brings together refugee leaders, service providers and community organisations across the two metropolitan areas where we will be recruiting. Additionally, as we have done in a previous study among refugees located in the same setting,^{2 17} the study team will work with community partners and religious leaders to recruit participants from community settings such as worship centres (eg, churches and mosques) and community centres. We will leverage these relationships for community mobilisation and information dissemination about the study. Based on previous experience working with refugee communities, the first step to successful recruitment and retention is speaking to leaders and community members openly about the study so that they may understand the proposed study clearly prior to any recruitment. Therefore, in line with the a priori selected implementation strategies,¹⁰¹ before any recruitment efforts, the PI will conduct meetings with leaders and do presentations about the study in these community settings. Recruitment materials and information will be distributed at community gatherings, religious and community centres, and shops owned by resettled refugees. Moreover, we will share the same information through social media platforms such as Facebook posts and WhatsApp groups where many community members exchange important information. Interested parents and youth will contact the PI or designated research assistant for details. Verbal consent to be contacted by a research staff and to schedule a follow-up meeting will be obtained. Youth will be contacted for assent only after receiving consent from their parents.

Retention procedures

For accurate and up-to-date tracing purposes, participants' contact information, including addresses and telephone numbers, will be collected using a detailed future contact form. Additionally, we will collect contact information of close contacts of the participant who will always know how to reach the participant. We will make it clear to the participants that no other information will be discussed with their contacts other than to ask them to help the research team locate and get in touch with the

participant. We will encourage the participant to inform the contacts that they have provided their contact information to the research team in the event that it becomes necessary to reach them through these individuals in the future. More important, in line with CBPR principles,^{99 100} to facilitate community engagement and increase the likelihood of acceptability of the intervention, research assistants will be recruited from the community and will stay in constant contact with participants during the intervention. We will leverage this avenue of communication to retain high participation rates in the study. We will keep records on drop-out and test for attrition bias based on data we will have prior to study drop-out. We will also learn from participants through qualitative interviews to learn about implementation and retention strategies in preparation for larger R01 trials to test the effectiveness.

Patient and public involvement

As described above, this study employs a CBPR approach.^{99 100} In this regard, from the design of the study, we consulted community and faith leaders, as well as non-profit agencies working with resettled refugee communities, to help identify areas of need and provide letters of support to help with recruitment of study participants. Community and religious leaders have taken the responsibility to share information with their members and have helped organise participant recruitment drives. Additionally, faith and community organisations are providing office space for intervention delivery and all the research assistants collecting data and delivering the intervention are recruited from the community. At the conclusion of the study, we will plan a dissemination conference where key stakeholders will be invited to hear the study findings. Community members and the research assistants will be heavily involved in the planning of the dissemination conference.

Potential challenges and alternative strategies

We do not anticipate major threats to study implementation, but we recognise potential concerns. Regarding study retention and completion, we have a stringent retention plan that includes virtual plans for session attendance for the peer mentorship component of the intervention. We expect to achieve enrolment goals and retention rates of about 80% based on our previous and ongoing studies in other equally challenging settings. Should recruitment, enrolment or retention deviate from anticipated rates, we will: (1) go to the youth's last known home address (indicated on the future contact form) to follow up with them; (2) work with community organisations, religious institutions and community leaders to obtain the current contact information (including workplace/telephone numbers); (3) work with research assistants who are in constant communication with the participants to obtain participants' contact and (4) provide incentives such as a series of badges, token recognition items, rewards, privileges and experiences systematically integrated into intervention activities.

Quantitative data collection

Assessments will take place at baseline, 16 weeks and 6 months postintervention. The data will be collected by trained research assistants in a private location at community centres or in the participants' homes (if privacy is assured)—and will take approximately 45–60 min. All interviewers will be recruited from the respective communities and will receive a highly structured and intensive training conducted by the principal investigator (PI). For questions measuring sensitive behaviours (eg, risk taking, substance use), we will use audio computer-assisted self-interviews where the participant takes the survey himself/herself on a tablet/iPad. Non-sensitive questions will be interviewer-administered. As needed, we will translate

and back-translate instruments to languages spoken in the target population. Data collection tools are listed in [table 2](#).

Qualitative data collection (aims 1–3)

To assess the acceptability of the intervention, we will use the CSQ-8.¹⁰⁹ This will be adapted to the RRF4H study, and questions will include seeking information about how helpful the RRF4H is in addressing mental health symptoms and how likely the participants are to recommend the programme to others in their network. Participants will be asked for feedback on intervention acceptability and relevance via semi-structured interviews on intervention completion. We will randomly select 20 participants (n=10 participants

Table 2 Variables, instruments, reliability and assessment time points

Variable	Measurement	Respondent	Time point
Demographics			
Age; socioeconomic status; gender	Sociodemographic questionnaire	Child/parent	B, 16W, 6M
Secondary outcomes and potential mechanisms of change			
Family/social support	Family cohesion ¹¹² ; Multidimensional scale of perceived social support ¹³⁷ ; UCLA Loneliness Scale Version 3 ¹³⁸	Child/parent	B, 16W, 6M
Family communication	Child-parent Communication ¹³⁹	Child/parent	B, 16W, 6M
Family functioning	FAD ¹⁴⁰	Child/parent	B, 16W, 6M
Mental health stigma	Paediatric Self-Stigmatisation Scale ¹⁴¹	Child/parent	B, 16W, 6M
Self-efficacy	Tennessee Self-Concept Scale ¹⁴² ; General Self-Efficacy Scale ¹⁴³	Child/parent	B, 16W, 6M
Hopelessness	Beck Hopelessness Scale	Child/parent	B, 16W, 6M
Improved peer support/relationships	Strengths and Difficulties Questionnaire (SDQ) ¹⁴⁴	Child	B, 16W, 6M
Improved prosocial attitudes/conduct problems	SDQ ¹⁴⁴	Child	B, 16W, 6M
Exposure to potentially traumatic event (PTE)	The Child PTSD Symptoms Scale for DSM-V Self-Report (CPSS-V SR) ¹⁴⁵ ; Harvard Trauma Questionnaire (HTQ) ¹⁴⁶	Child/parent	B, 16W, 6M
Adverse childhood experiences	Adverse Childhood Experience scale ¹⁴⁷	Child only	B, 16W, 6M
Risk-taking behaviours (substance use, sexual risk-taking behaviours)	Youth Risk Behaviour Survey ¹⁴⁸	Child only	B, 16W, 6M
Primary outcomes			
Post-traumatic stress disorder (PTSD)	The CPSS-V SR ¹⁴⁵ ; HTQ ¹⁴⁶	Child/parent	B, 16W, 6M
Depression and anxiety	Hopkins Symptoms Checklist-37 ¹⁴⁹ ; Hopkins Symptoms Checklist-25 ¹⁵⁰	Child/parent	B, 16W, 6M
Aggressive behaviour, antisocial behaviour	Child Behaviour Checklist for ages 6–18 years ¹⁵¹	Child only	B, 16W, 6M
Intervention feasibility, acceptability and fidelity			
Intervention feasibility	Recruitment rates; proportion eligible and enrolled ($\geq 70\%$ enrolment considered feasible) ¹¹⁵	Child/parent	B, 16W, 6M
Intervention acceptability	Client Satisfaction Questionnaire-8 (to be adapted) ¹⁰⁹	Child/parent	6M
Implementation fidelity	Observation ¹⁵²	Researcher	Ongoing

B, baseline; DSM-V, Diagnostic and Statistical Manual of Mental Disorders-5; FAD, Family Assessment Device; M, months; UCLA, University of California, Los Angeles; w, weeks.

per arm) to participate in semistructured interviews. A stratified purposeful sampling strategy¹¹⁰ will be used to randomly select participants from the highest (n=5) and lowest quartiles (n=5) on the primary outcomes. The literature shows that saturation is achieved between 12 and 16 interviews.^{111 112} Hence, both the within-group and overall sample size will be sufficient for theoretical saturation. This sampling method will ensure that participants with varying experiences are represented. The study will examine barriers to and facilitators of the implementation of the RRF4H intervention, applying the EPIS framework.⁵⁶ Specifically, semistructured interviews will: (1) explore participants' intervention experiences and perceived impact of the intervention on mechanisms of change; (2) identify multilevel facilitators and barriers to intervention implementation and participation and (3) explore participants' perceptions of the implementation strategies used and any modifications that were necessary throughout the course of the study. Any modifications to the implementation strategies will be carefully documented as proposed by Haley *et al.*¹⁰³ and Miller *et al.*¹⁰⁴ The interviews will provide in-depth data on participants' processes and experiences with the intervention; processes behind key outcomes, mechanisms of change and mediating variables; and potential individual, family, contextual, programmatic, and structural factors affecting their experiences. Each interview will last about 45–60 min and will be audio taped. In addition, all research assistants from the community will participate in a 2-hour focus group discussion to provide their experiences with the intervention and implementation. Study data were collected and managed using Research Electronic Data Capture (REDCap) electronic data capture tools hosted at Washington University in St. Louis.^{113 114} REDCap is a secure, web-based software platform designed to support data capture for research studies, providing (1) an intuitive interface for validated data capture; (2) audit trails for tracking data manipulation and export procedures; (3) automated export procedures for seamless data downloads to common statistical packages and (4) procedures for data integration and interoperability with external sources. The REDCap mobile application allows for data collection without the need for internet access, and all the data collected are then uploaded to the electronic database on connection of the device to the internet.

Analysis for aim 1

Intervention feasibility

Binary feasibility metrics will include enrolment of 70% or more of potential participants approached and retention of 80% or more participants at 12 months. Based on previous studies, attainment of 70% or above enrolment rate will indicate feasibility of the study.¹¹⁵

Intervention acceptability

We will determine participant satisfaction with the study based on the CSQ-8 (to be adapted).¹⁰⁹ The score ranges from 8 to 32, with higher scores indicating greater satisfaction with the study. The scores will be grouped into four categories: dissatisfied (8–21); mildly satisfied (22–25); satisfied (26–30) and very satisfied (31–32).

Data analysis for aim 2

Primary descriptive analyses

Frequency tables for all variables and measures of central tendency and variability for continuous variables will characterise the sample overall for biological parents and their children.

Primary inferential analyses

Participants will be randomly assigned to control or treatment group at the community level so that all participants enrolled at a particular community receive the same intervention. Based on our power analysis, we will have a sufficient sample size to initially test several hypotheses to assess preliminary efficacy of the combination intervention (RRF4H). We, therefore, offer the following hypotheses to assess the preliminary efficacy of RRF4H versus usual care: relative to adolescents in the usual care group, adolescents in the RRF4H group will have lower mean scores of PTSD (H1), depression (H2), anxiety (H3), aggressive behaviour (H4) and antisocial behaviour (H5). To test these hypotheses, we will fit two-level generalised linear mixed models (GLMMs) to these continuous primary outcomes. The models will include fixed effects for study arm, time and their interactions. To maximise rigour our analyses will follow an intention-to-treat (ITT) approach such that all adolescents will be included in the analysis irrespective of whether they have complete outcome data. As described above, likelihood-based and multiple imputation (MI) methods will be used to address missing data with sensitivity analyses being performed with pattern-based MI to examine the robustness of results under different missing data mechanism assumptions. To test H1–H5, we will perform time-averaged comparisons of post-baseline repeatedly measured observations across study arms to examine intervention effects over the duration of the postintervention study period at alpha (α)=0.05. Any additional post hoc comparisons (eg, paired comparisons of intervention groups at each time point) will maintain a nominal α of 0.05 through the use of simulation-based stepdown multiple comparison methods.¹¹⁶ To maximise the rigour of these analyses, quasi-likelihood estimation methods will not be used because simulation studies show they can yield biased results for binary outcome data with few observations per cluster.¹¹⁷ Instead, GLMMs will be estimated by maximum likelihood via adaptive Gaussian quadrature. To promote stable results, a minimum of 15 numerical integration points will be used to obtain final estimates.

Data analysis for aim 3

Qualitative data analysis

The primary focus of this specific aim is on understanding barriers and facilitators to implementation and implementation strategies. Thus, we will apply a more integrative approach, using both inductive and deductive approaches to delineate these barriers and facilitators. Specifically, guided by the grounded theory—and integrative theory that combines both deductive and inductive approaches to describe a social phenomenon—we will write analytical memos to further develop categories, themes and subthemes, and employ axial coding to link categories to one another^{118 119} and integrate the ideas that emerge from the data.^{120–122} Using analytical induction techniques,^{119 123} transcripts will be read multiple times for initial coding. For initial coding, randomly selected 10 interview transcripts will be read multiple times and independently coded using a priori (ie, from the interview guide) or emergent themes (also known as open coding).^{120 124} Broader themes/categories will be broken down into smaller, more specific units until no further subcategory is necessary. For example, potential themes and subthemes on intervention experiences may include barriers and facilitators for programme participation at the individual level (eg, motivation, readiness to change, household income), family level (eg, competing demands, support); programme level (eg, content relevance) and macro level (eg, cultural norms).

Data integration

We will employ mixed methods in our data analysis; therefore, findings from the quantitative and qualitative analysis will need to be integrated in data interpretation and discussion.¹²⁵ We will use these qualitative data to provide additional explanation and context for our quantitative findings. These findings will also further contextualise the mediators and moderators and may identify additional variables under consideration as mechanisms of change. Moreover, the qualitative findings will: (1) complement and expand on our understanding of attendance and participant satisfaction (also measured quantitatively) for the intervention group and (2) inform our overall understanding of the impact of the intervention by providing insights to implementation processes, acceptability and feasibility.

Sample size and power calculations

The primary purpose of this study is to test the feasibility and acceptability of the intervention and not to conduct formal hypothesis testing. However, we conducted sample size and power analysis using G*Power V.3.1.9.7 Software¹²⁶ to compare mean differences for the emotional and behavioural health outcomes between two independent groups (treatment and control group). We set power at 0.80, alpha=0.05 and medium effect size of 0.5.¹²⁷ These assumptions yielded a minimum sample size of 128, that is, 64 per study arm to detect statistically

meaningful difference between the intervention and control groups. Allowing for a conservative attrition rate of 20% at 6 months follow-up, we propose to recruit 20% above the minimum sample size, yielding a total sample size of 154, that is, 77 participants per study arm.

Data management responsibilities and records retention

Data collection will be the responsibility of the clinical trial staff at the site under the supervision of the principal investigator. The investigator will be responsible for ensuring the accuracy, completeness, legibility, security and timeliness of the data reported. Additionally, as part of the qualitative data collection, this study involves making audio recordings of participants. The recordings will be made during qualitative data interviews in order to capture all the information participants provide. The recordings will then be transcribed to be analysed. If the responses are in a language other than English, the transcripts will be translated to English and uploaded to qualitative data analysis software for coding and analysis. Only the research team will have access to these recordings and transcripts, which will be destroyed 6 years after the conclusion of the study.

All source documents such as consent forms will be completed in a neat, legible manner to ensure accurate interpretation of information. Hard copies of the study visit worksheets will be provided for use as source document worksheets for recording data for each participant consented/enrolled in the study. Data recorded in the electronic case report form derived from source documents will be consistent with the data recorded on the source documents.

To protect the integrity of the participants' data, the following procedures will be followed. First, the data collected from the study participants will be used only for the purpose of research. All data will be kept confidential. We will not share any information or answers we get from the participants with their families, friends or peer mentors. Since the study is a longitudinal one, we will maintain lists of participants with links between identifying information and code numbers. Only the PI will have access to these lists, which are kept in password protected electronic files. Other study personnel have access to individual participants' names and code numbers on a need-to-know basis in order to adequately perform their duties. For example, interviewers enter the correct code number of the participant whom they are interviewing to each individual electronic file. Each research assistant is assigned a unique login ID and password, with varying levels of data access; this allows tracking of who accesses data, or entered individual data, for auditing and data security purposes. All personnel must complete certain levels of training before they are granted access to this identifying information. They must complete the Human Subjects Protection Training and Good Clinical Practice. Personnel also sign confidentiality statements that specify that

if the participants' confidentiality is breached unintentionally, personnel will follow the procedures for reporting this breach to the PI. The confidentiality statements also state that unintentional or deliberate violations of participants' confidentiality may result in demotion or termination depending on the severity of the event. The project personnel will also participate in training with the PI regarding data safety, confidentiality of participants, limits of confidentiality and proper administration of the study protocol.

All requests, current and future, to use the data will be reviewed by the PI. Any data files that are provided to other individuals are stripped of identifiers and contain only code numbers so that data across multiple assessment waves can be matched. During the informed consent/assent process, participants will be notified of the above procedures. Study documents will be destroyed 6 years after the conclusion of the study.

Missing data

We will train all research assistants to ensure that missing data is minimised by checking assessments for missing data before they leave the participants. Importantly, as data will be collected using electronic devices, to minimise missing data on critical responses, we will implement validation procedures so that a research assistant cannot move to the next question without entering a response to a question. Additionally, we will use MIS IDA Q^{128 129} to check for data-entry errors and missing values.¹²⁹ Frequency tables for all variables and measures of central tendency and variability for continuous variables will characterise the sample overall and by randomisation group. If the groups differ significantly at baseline on one or more covariates, we will use methods based on the Rubin causal model (eg, propensity scores, double-robust estimation) to obtain the desired effect estimates under the counterfactual assumption of balanced groups.^{130 131} We will address incomplete data with direct maximum likelihood (ML) and MI¹³² because they make the relatively mild assumption that incomplete data arise from a conditionally missing-at-random (MAR) mechanism.¹³³ Auxiliary variables will be included to help meet the MAR assumption¹³⁴ and sensitivity analyses will be conducted with pattern-based MI to assess the robustness of the MAR assumption.¹³⁵ The proposed analyses will be conducted using validated algorithms in SAS.¹³⁶ All programme code and results will be documented extensively and archived to enable future review, transparency and results reproducibility.

Data safety monitoring

An independent study-specific Data Safety Monitoring Board (DSMB) will be appointed comprising four members. Together, the DSMB members represent the appropriate expertise (biostatistics, community engagement, refugee research, adolescent psychiatry and clinical trials) and will provide oversight to assure that the trial accrues at a sufficient rate, and that the safety and

privacy of all study participants are assured. The DSMB will meet initially every quarter (3 months), then later at a frequency of 3–6 months as determined by DSMB members, to review study progress and to monitor adverse events (AEs) or more often if deemed necessary by the committee. The DSMB will review the results of a single interim analysis, performed after 50% of the sample size has completed the study (exact timing at the discretion of the DSMB).

AE reporting

This protocol uses the definition of AE from 21 CFR 312.32 (a): any untoward medical occurrence associated with the use of an intervention in humans, whether or not considered intervention related. A serious AE (SAE) is any AE, that is: fatal, life-threatening, requires or prolongs hospital stay, results in persistent or significant disability or incapacity, suicide attempt or ideation, and an important medical event. Important medical events are those that may not be immediately life-threatening but are clearly of major clinical significance. They may jeopardise the subject and may require intervention to prevent one of the other serious outcomes noted above. For example, drug overdose or abuse, a seizure that did not result in inpatient hospitalisation, or intensive treatment of bronchospasm in an emergency department (ED) would typically be considered serious. All AEs that do not meet any of the criteria for SAEs should be regarded as non-SAEs.

Consistent with NIH policy, the PI is responsible for the management of participants and accurate written documentation, investigation and follow-up of all possible study-related AEs. The PI is also responsible for describing and adhering to the procedures for identifying, monitoring and reporting any reportable events, including AEs, SAEs, death and unanticipated problems. If the SAE/AE is observed by a research assistant, he/she will contact the PI immediately (no later than 12 business hours). All documentation for the events listed above will be made in writing to the WUSTL IRB and NIMH Programme Official (PO) by the PI. The PI will be responsible for filing the paperwork/report related to the reporting of SAE/AE to the above-mentioned institutions within the timeline specified in the table.

Documentation will include identifying information for the research protocol (eg, the investigator's name, project title, the grant/contract number), the date the event occurred and the date the study team became aware of the event, a detailed description of the event and impact on the participants, a detailed description of measures taken, confirmation that the appropriate monitoring entities and regulatory bodies have been notified. As needed, a description of any changes to the protocol or other corrective actions that have been taken or will be undertaken in response to the event. The following information about events will be collected: (a) the onset and resolution of the event; (b) an assessment of the severity or intensity of the AE; (c) an assessment of the relationship of the event to the intervention (definitely,

probably, possibly or not related) and (d) action taken (eg, none, discontinuation of participation in the study). The PI will review all events to determine the severity of the event, assign attribution to the event and monitor the event until its resolution. In consultation with the PI, a trained member of the study team will be responsible for conducting an evaluation of an SAE and shall report the results of such evaluation to the NIH and the reviewing Institutional Review Board (IRB) as soon as possible, but in no event later than 10 working days after the investigator first learns of the event.

ETHICS AND DISSEMINATION

All study procedures received approval from Washington University in St. Louis IRB (IRB #202307081). Voluntary written informed assent and consent will be obtained from all participants, adolescents and their parents, respectively, by trained research assistants or the principal investigator. To avoid coercion, assenting and consenting process and interviews will be conducted separately for adolescents and their parents once the parents provide consent for their child/children to participate. Given the complex nature of the study, the consenting process will be conducted in the language that the participant is most comfortable speaking. Importantly, as interview questions or discussions during the intervention may elicit emotional reactions from participants (both youth and their parents), the study team has partnered with an independent clinical mental health therapist licensed to practise in the state of Nebraska to be available for any assistance if any participants should experience emotional reactions. Participants who require urgent medical attention will be taken to the ED for care and connected to resources within their community for additional services that they can continue to use beyond the study period.

To our knowledge, this is the first RCT designed to address emotional and behavioural health outcomes related to intergenerational trauma among US-born children of parents resettled as refugees. This is crucial because children belonging to resettled refugee families are at elevated risk of developing serious mental health challenges compared with their US native peers. Therefore, the research team will facilitate learning across stakeholders and maximise use of the evidence generated through dissemination meetings. Specifically, we will engage key stakeholders, including community and religious leaders, parents or guardians of students participating in the study, state and local public health agencies and school personnel to disseminate critical information about the study. We will produce policy briefs and infographics to disseminate information related to the study tailored to different stakeholders. We will attend community meetings and use novel approaches such as use of WhatsApp and Facebook messaging, widely used in resettled refugee communities, for information dissemination. We are also planning to organise a dissemination conference at the end of the study period, at which various stakeholders and participants will be invited to hear the findings of the study. We

will also disseminate study findings through publications in peer-reviewed scientific journals and both national and international conferences.

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Acknowledgements We appreciate the support of staff members and volunteers at the International Center for Child Health and Development (ICHAD) for the study implementation. We are incredibly grateful to all the community partners, including the Omaha Refugee Task Force (ORTF), New Family Alliance, the First Presbyterian Church in Lincoln, Nile Lutheran Church, Rev. James Kuajien Deng, Mr Mahmed Jimale of East African Development Association of Nebraska, Mr Duol Rut of Turning Point Psychotherapy, Mr Kor Tot Rualmim, Mr Keak Lam Kierkok and Ms Nyabuay Wuol of International Council for Refugees and Immigrants for their assistance in community engagement and recruitment of participants; Mr Billy Rucker, Dr. Eboni Sterling and Southern Illinois University Edwardsville (SIUE) for their assistance in design and adaptation to virtual environment of TeenAge Health Consultants (TAHC) peer mentorship program.

Contributors NTT is the guarantor author. NTT conceptualised the manuscript and wrote the original draft of the manuscript; TSB contributed to the original draft and reviewed the final draft of the manuscript; TWL contributed to literature search and manuscript drafting; BJP provided substantial guidance in the conceptualisation of the implementation science component of the intervention; SG helped conceptualise the statistical analysis and reviewed the manuscript; MM provided guidance on the development of the clinical components of the study protocol; and FMS provided overall guidance on the conceptualisation and development of the entire manuscript. All authors read and approved the final manuscript.

Funding This research was funded by the National Institute of Mental Health (NIMH) (Grant: K01MH131872; PI: NTT).

Disclaimer The content is solely the responsibility of the authors and does not necessarily represent the official views of the NIMH. The funder had no role in the design of this study and did not have any role during its execution, analyses, interpretation or submission of results.

Competing interests None declared.

Patient and public involvement Patients and/or the public were involved in the design, or conduct, or reporting, or dissemination plans of this research. Refer to the Methods section for further details.

Patient consent for publication Not applicable.

Provenance and peer review Not commissioned; peer reviewed for ethical and funding approval prior to submission.

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